# 3D Volumetric Brain MRI Glioma Segmentation

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## 1. PROJECT OVERVIEW & SCOPE

Volumetric segmentation of glioblastoma sub-regions using 3D U-Net deep neural network on multi-modal FLAIR and T1ce MR neuro-imaging scans.

## 2. VOLUMETRIC SEGMENTATION METRICS

Dice similarity coefficient of 91.2% for whole tumor and 88.5% for active enhancing tumor core across BraTS clinical cohorts.

## 3. CLINICAL RELEVANCE

Provides automated volumetric measurements for radiation oncology planning and surgical resection guidance.